

ATKIN, MN, USA

WMO: 727504

Lat: 46.548N

Lon: 93.677W

Elev: 1204

StdP: 14.07

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.1	-14.9	-27.9	1.2	-19.0	-21.9	1.7	-11.5	18.5	17.5	16.7	17.5	1.7	250	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	85.8	74.4	82.4	71.6	81.0	69.9	76.8	83.4	74.4	81.0	71.9	78.5	9.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	136.0	81.7	72.4	125.3	79.6	70.0	115.6	77.1	41.1	83.5	38.8	80.4	36.4	77.2	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	16.1	13.8	-28.0	90.6	7.1	2.5	-33.1	92.4	-37.2	93.9	-41.2	95.3	-46.4	97.2
			-28.2	80.1	6.8	2.5	-33.1	81.9	-37.1	83.3	-40.9	84.8	-45.9	86.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.0	10.4	14.7	27.7	40.9	52.9	62.5	67.4	64.5	57.2	44.6	30.1	17.1
	DBStd	22.06	14.21	13.21	12.81	9.38	8.09	6.90	6.08	5.89	8.59	8.83	10.67	12.22
	HDD50	5135	1227	988	695	298	59	3	0	0	28	217	599	1021
	HDD65	9040	1692	1408	1156	724	386	125	44	80	259	634	1046	1486
	CDD50	1841	0	0	4	24	149	378	539	450	245	49	3	0
	CDD65	271	0	0	0	1	10	49	118	65	27	1	0	0
	CDH74	2991	0	0	0	11	195	568	1258	708	236	15	0	0
	CDH80	667	0	0	0	1	43	115	324	143	40	1	0	0
	WSAvg	5.6	5.4	5.7	6.1	6.8	6.4	5.3	4.6	4.4	5.3	6.0	6.0	5.5
	PrecAvg	28.7	0.8	0.7	1.3	2.3	3.1	4.4	4.4	3.3	3.0	3.1	1.5	0.9
(15) Precipitation	PrecMax	33.0	2.1	1.7	3.2	6.5	6.1	7.4	8.0	6.3	5.9	6.7	3.8	1.7
	PrecMin	19.6	0.2	0.1	0.6	0.6	1.3	1.5	1.9	1.4	0.9	0.7	0.1	0.3
	PrecStd	3.5	0.5	0.4	0.7	1.2	1.2	1.5	1.5	1.5	1.3	1.6	1.0	0.4
	PrecStd	3.5	0.5	0.4	0.7	1.2	1.2	1.5	1.5	1.5	1.3	1.6	1.0	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	47.8	65.6	74.9	84.5	87.7	90.4	88.3	84.0	75.3	63.5	45.4
		MCWB	37.8	42.5	54.6	57.8	67.4	74.1	78.0	76.4	73.9	61.5	53.5	40.4
	2%	DB	36.9	42.3	56.8	69.6	79.5	82.5	86.4	83.7	80.8	70.1	55.3	39.3
		MCWB	34.0	37.5	47.4	54.2	63.2	70.9	75.8	73.4	70.2	60.4	47.8	35.7
	5%	DB	34.0	37.1	50.3	63.7	74.7	80.8	83.5	81.1	75.5	63.9	50.3	36.4
		MCWB	31.8	34.0	44.1	50.2	59.9	68.6	74.3	70.3	66.7	55.1	44.8	34.1
	10%	DB	30.0	34.2	45.1	59.0	70.2	77.1	81.2	78.7	72.6	59.6	45.7	34.0
		MCWB	28.8	31.8	39.9	47.5	57.4	65.9	71.3	68.5	64.7	52.6	41.6	32.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.8	42.1	57.2	61.1	73.3	77.2	81.2	78.8	75.3	66.7	55.3	40.8
		MCDB	43.2	47.3	62.7	69.9	79.7	84.2	86.8	84.9	82.1	70.7	60.4	44.6
	2%	WB	34.4	38.0	49.5	55.3	66.4	74.1	78.1	75.1	71.8	60.6	49.0	36.9
		MCDB	36.9	41.2	55.7	65.3	74.8	80.7	84.3	81.9	78.1	67.3	55.4	39.4
	5%	WB	31.6	34.8	44.6	51.9	62.4	70.9	75.3	72.2	68.8	57.0	45.3	33.9
		MCDB	33.4	37.2	49.9	62.4	70.7	77.7	81.7	78.7	74.6	64.0	49.1	36.2
	10%	WB	28.4	31.6	39.9	48.3	59.3	67.9	72.8	69.8	66.2	53.3	41.5	31.7
		MCDB	29.5	34.0	45.2	57.9	67.8	74.8	79.4	76.1	71.3	58.2	46.1	33.7
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	20.5	20.4	22.4	24.3	22.8	22.6	23.1	22.4	19.1	16.1	15.6
		MCDBR	18.2	20.4	25.8	32.0	32.1	26.4	24.4	25.7	25.6	26.8	23.9	17.5
		MCWBR	15.6	16.2	17.3	19.0	18.2	16.3	15.8	16.8	17.2	17.2	17.0	14.5
	5% WB	MCDBR	17.0	18.4	24.3	28.7	26.3	22.7	22.8	23.1	22.2	23.0	21.8	16.6
		MCWBR	14.9	14.8	17.2	18.5	17.4	15.9	16.3	16.7	16.2	16.3	16.5	14.1
(40) Clear-Sky Solar Irradiance	taub		0.251	0.270	0.307	0.361	0.382	0.377	0.379	0.387	0.358	0.310	0.285	0.260
	taud		2.385	2.340	2.380	2.336	2.331	2.372	2.390	2.382	2.438	2.567	2.512	2.408
	Ebn at Noon		274	289	292	284	280	282	279	272	271	273	259	256
	Edh at Noon		22	29	32	37	39	37	36	35	30	23	20	19
(44) All-Sky Solar Radiation	RadAvg		501	856	1209	1486	1694	1844	1963	1643	1238	759	471	353
	RadStd		55	76	117	141	127	124	80	129	92	104	48	35

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page